

SCOPE: Algorithms for each causal learner

Algorithm 1 SCOPE-S: Training and Inference

Training (Offline)

Input: Dataset $\mathcal{D} = \bigcup_{c \in C} \bigcup_{k \in \mathcal{K}^{(c)}} \left\{ (\sigma_k^{(c)}, a_k^{obs, (c)}, y^{(c)}) \right\}$, action spaces $\{A_k\}_{k=1}^K$

- 1: $\hat{V}_{K+1}^{(c)} = y^{(c)} \quad \forall c \in \{1, \dots, C\}; \mathbf{M} = []$ ▷ Initialize
- 2: **for** $k = K, K-1, \dots, 1$ **do** ▷ Iterate backwards
- 3: $\mathcal{C}_k = \{c | k \in \mathcal{K}^{(c)}\}$ ▷ Identify cases that visited k
- 4: $\mathcal{M}_k \leftarrow \text{Regress } \hat{V}_{k+1}^{(c)} \text{ on } (\sigma_k^{(c)}, a_k^{obs, (c)}) \text{ for all } c \in \mathcal{C}_k$ ▷ S-learner Q-function
- 5: Append $\mathcal{M}_k$ to $\mathbf{M}$
- 6: **for** each case $c \in \mathcal{C}_k$ **do**
- 7: $\hat{a}_k^{opt, (c)} = \text{argmax}_{a_k \in A_k} \mathcal{M}_k(\sigma_k^{(c)}, a_k)$ ▷ Estimated optimal action
- 8: $\hat{Q}_k^{obs, (c)} = \mathcal{M}_k(\sigma_k^{(c)}, a_k^{obs, (c)})$ ▷ Q-value of observed action
- 9: $\hat{Q}_k^{opt, (c)} = \mathcal{M}_k(\sigma_k^{(c)}, \hat{a}_k^{opt, (c)})$ ▷ Q-value of optimal action
- 10: $\hat{V}_k^{(c)} = \hat{V}_{k+1}^{(c)} + \hat{Q}_k^{opt, (c)} - \hat{Q}_k^{obs, (c)}$ ▷ Regret-based value update
- 11: **end for**
- 12: **for** each case $c \notin \mathcal{C}_k$ **do**
- 13: $\hat{V}_k^{(c)} = \hat{V}_{k+1}^{(c)}$ ▷ Carry over value if point k is skipped
- 14: **end for**
- 15: **end for**
- 16: **return** $\mathbf{M}$

Inference (Online)

Input: Models $\{\mathcal{M}_k\}_{k=1}^K$, initial prefix $\sigma_{k_{first}}^{(c)}$

- 17: **while** case c is not completed **do**
 - 18: **if** case c reaches DECISION POINT $k \in \{1, \dots, K\}$ **then**
 - 19: Observe current prefix $\sigma_k^{(c)}$
 - 20: $\hat{a}_k^{opt, (c)} = \text{argmax}_{a_k \in A_k} \mathcal{M}_k(\sigma_k^{(c)}, a_k)$
 - 21: Execute action $\hat{a}_k^{opt, (c)}$
 - 22: **end if**
 - 23: **end while**
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Algorithm 2 SCOPE-T: Training and Inference

Training (Offline)

Input: Dataset $\mathcal{D} = \bigcup_{c \in \mathcal{C}} \bigcup_{k \in \mathcal{K}^{(c)}} \left\{ (\sigma_k^{(c)}, a_k^{obs, (c)}, y^{(c)}) \right\}$, action spaces $\{A_k\}_{k=1}^K$

- 1: $\hat{V}_{K+1}^{(c)} = y^{(c)} \forall c \in \{1, \dots, C\}$; $\mathbf{M} = []$ ▷ Initialize
- 2: **for** $k = K, K-1, \dots, 1$ **do** ▷ Iterate backwards
- 3: $\mathcal{C}_k = \{c \mid k \in \mathcal{K}^{(c)}\}$ ▷ Identify cases that visited k
- 4: $\mathcal{M}_k = []$ ▷ Fit a separate model for each action
- 5: **for** each action $a_k \in A_k$ **do**
- 6: $\mathcal{M}_k[a_k] \leftarrow \text{Regress } \hat{V}_{k+1}^{(c)} \text{ on } \sigma_k^{(c)} \text{ for } c \in \mathcal{C}_k \text{ s.t. } a_k^{obs, (c)} = a_k$ ▷ T-learner Q-function
- 7: **end for**
- 8: Append $\mathcal{M}_k$ to $\mathbf{M}$
- 9: **for** each case $c \in \mathcal{C}_k$ **do**
- 10: $\hat{a}_k^{opt, (c)} = \text{argmax}_{a_k \in A_k} \mathcal{M}_k[a_k](\sigma_k^{(c)})$ ▷ Estimated optimal action
- 11: $\hat{Q}_k^{obs, (c)} = \mathcal{M}_k[a_k^{obs, (c)}](\sigma_k^{(c)})$ ▷ Q-value of observed action
- 12: $\hat{Q}_k^{opt, (c)} = \mathcal{M}_k[\hat{a}_k^{opt, (c)}](\sigma_k^{(c)})$ ▷ Q-value of optimal action
- 13: $\hat{V}_k^{(c)} = \hat{V}_{k+1}^{(c)} + \hat{Q}_k^{opt, (c)} - \hat{Q}_k^{obs, (c)}$ ▷ Regret-based value update
- 14: **end for**
- 15: **for** each case $c \notin \mathcal{C}_k$ **do**
- 16: $\hat{V}_k^{(c)} = \hat{V}_{k+1}^{(c)}$ ▷ Carry over value if point k is skipped
- 17: **end for**
- 18: **end for**
- 19: **return** $\mathbf{M}$

Inference (Online)

Input: Model lists $\{\mathcal{M}_k\}_{k=1}^K$, initial prefix $\sigma_{k_{first}}^{(c)}$

- 20: **while** case c is not completed **do**
- 21: **if** case c reaches DECISION POINT $k \in \{1, \dots, K\}$ **then**
- 22: Observe current prefix $\sigma_k^{(c)}$
- 23: $\hat{a}_k^{opt, (c)} = \text{argmax}_{a_k \in A_k} \mathcal{M}_k[a_k](\sigma_k^{(c)})$
- 24: Execute action $\hat{a}_k^{opt, (c)}$
- 25: **end if**
- 26: **end while**

Algorithm 3 SCOPE-RA: Training and Inference

Training (Offline)

Input: Dataset $\mathcal{D} = \bigcup_{c \in C} \bigcup_{k \in \mathcal{K}^{(c)}} \left\{ (\sigma_k^{(c)}, a_k^{obs, (c)}, y^{(c)}) \right\}$, action spaces $\{A_k\}_{k=1}^K$

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1:  $\hat{V}_{K+1}^{(c)} = y^{(c)} \forall c \in \{1, \dots, C\}$ ;  $\mathbf{M} = [ ]$ ;  $\mathbf{E} = [ ]$  ▷ Initialize
2: for  $k = K, K-1, \dots, 1$  do ▷ Iterate backwards
3:    $\mathcal{C}_k = \{c | k \in \mathcal{K}^{(c)}\}$  ▷ Identify cases that visited  $k$ 
4:    $\mathcal{M}_k \leftarrow \text{Regress } \hat{V}_{k+1}^{(c)} \text{ on } (\sigma_k^{(c)}, a_k^{obs, (c)}) \text{ for all } c \in \mathcal{C}_k$  ▷ S-learner to estimate Q-function
5:   Append  $\mathcal{M}_k$  to  $\mathbf{M}$ 
6:   for each case  $c \in \mathcal{C}_k$  do
7:     for each action  $a_k \in A_k$  do
8:        $\Phi_k^{(a_k)(c)} \leftarrow \text{Compute pseudo-outcome using } (\sigma_k^{(c)}, \mathcal{M}_k, a_k)$ 
9:     end for
10:  end for
11:   $\mathcal{E}_k \leftarrow \text{Regress } \Phi_k^{(a_k)(c)} \text{ on } (\sigma_k^{(c)}, a_k) \text{ for all } c \in \mathcal{C}_k$  ▷ Model to estimate causal effects
12:  Append  $\mathcal{E}_k$  to  $\mathbf{E}$ 
13:  for each case  $c \in \mathcal{C}_k$  do
14:     $\hat{a}_k^{opt, (c)} = \text{argmax}_{a_k \in A_k} \mathcal{E}_k(\sigma_k^{(c)}, a_k)$  ▷ Estimated optimal action
15:     $\hat{Q}_k^{obs, (c)} = \mathcal{M}_k(\sigma_k^{(c)}, a_k^{obs, (c)})$  ▷ Q-value of observed action
16:     $\hat{Q}_k^{opt, (c)} = \mathcal{M}_k(\sigma_k^{(c)}, \hat{a}_k^{opt, (c)})$  ▷ Q-value of optimal action
17:     $\hat{V}_k^{(c)} = \hat{V}_{k+1}^{(c)} + \hat{Q}_k^{opt, (c)} - \hat{Q}_k^{obs, (c)}$  ▷ Regret-based value update
18:  end for
19:  for each case  $c \notin \mathcal{C}_k$  do
20:     $\hat{V}_k^{(c)} = \hat{V}_{k+1}^{(c)}$  ▷ Carry over value if point  $k$  is skipped
21:  end for
22: end for
23: return  $\mathbf{E}$  (and  $\mathbf{M}$ )
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Inference (Online)

Input: Causal effect models $\{\mathcal{E}_k\}_{k=1}^K$, initial prefix $\sigma_{k_{first}}^{(c)}$

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24: while case  $c$  is not completed do
25:   if case  $c$  reaches DECISION POINT  $k \in \{1, \dots, K\}$  then
26:     Observe current prefix  $\sigma_k^{(c)}$ 
27:      $\hat{a}_k^{opt, (c)} = \text{argmax}_{a_k \in A_k} \mathcal{E}_k(\sigma_k^{(c)}, a_k)$ 
28:     Execute action  $\hat{a}_k^{opt, (c)}$ 
29:   end if
30: end while
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